

- 1) THIS IS AN IMPEDANCE CONTROLLED BOARD
- 2) TRACE WIDTH 9.5 MIL OUTER LAYERS, 4.4 MIL IN SIG2,SIG3,SIG4,SIG5 & SIG6 LAYERS TO BE 50 OHM +/-10% TOLERANCE.
- 3) TRACE WIDTH 9 MIL AND SPACING 18 MIL IN OUTER LAYERS TO BE 100 OHM +/-10 % TOLERANCE
- 4) TRACE WIDTH 4 MIL AND SPACING 8 MIL IN SIG2,SIG3,SIG4 & SIG6 LAYERS TO BE 100 OHM +/-10% TOLERANCE
- 5) TRACE WIDTH 6 MIL AND SPACING 5 MIL IN TOP LAYER TO BE 90 OHM +/-10 % TOLERANCE
- 5) ALL VIAS ARE TENTED, UNLESS MASK IS OPENED IN GERBERS.
- 6) VENDOR MAY ADJUST SOLDERMASK WHEREVER SOLDER MASK PADS ARE THE SAME SIZE(1:1) AS PER THE MANUFACTURING CAPABILITIES AND ALL OTHER SOLDER MASK PADS SHALL NOT BE MODIFIED.
PROVIDED NO ADJACENT COPPER IS EXPOSED AND NO CONFLICT IS PRODUCED WITH ANY STATED VIA TENTING/COVERING REQUIREMENTS
- 7) ON PAD VIAS TO BE FILL AND FLAT WITH NON CONDUCTIVE MATERIAL OR PLUG IN REQUIRED.

Symbol	Quantity	Finished Hole Size	Plated	Hole Type	Drill Layer Pair	Tolerance
A	51	7.87mil (0.200mm)	PTH	Round	Top Layer - Bottom Layer	-3.00mil
B	458	8.00mil (0.203mm)	PTH	Round	Top Layer - Bottom Layer	-3.00mil
C	415	10.00mil (0.254mm)	PTH	Round	Top Layer - Bottom Layer	-3.00mil
D	159	12.00mil (0.305mm)	PTH	Round	Top Layer - Bottom Layer	-3.00mil
E	3	27.56mil (0.700mm)	PTH	Round	Top Layer - Bottom Layer	-3.00mil
F	4	33.47mil (0.850mm)	PTH	Round	Top Layer - Bottom Layer	-3.00mil
G	2	37.40mil (0.950mm)	NPTH	Round	Top Layer - Bottom Layer	-2.00mil
H	29	40.00mil (1.016mm)	PTH	Round	Top Layer - Bottom Layer	-3.00mil
P	210	40.16mil (1.020mm)	PTH	Round	Top Layer - Bottom Layer	-3.00mil
J	6	43.31mil (1.100mm)	NPTH	Round	Top Layer - Bottom Layer	-2.00mil
K	6	43.31mil (1.100mm)	PTH	Round	Top Layer - Bottom Layer	-3.00mil
L	4	59.06mil (1.500mm)	PTH	Round	Top Layer - Bottom Layer	-3.00mil
M	16	66.93mil (1.700mm)	PTH	Round	Top Layer - Bottom Layer	-3.00mil
N	4	125.98mil (3.200mm)	PTH	Round	Top Layer - Bottom Layer	-3.00mil
	1367 Total					

Layer	Name	Material	Thickness	Constant	Board Layer Stack
	Top Overlay				
	Top Solder	Solder Resist	2.00mil	3.5	
1	Top Layer		1.88mil		
	Dielectric 1	ISOLA 370 HR	6.05mil	4.2	
2	GND1/HV		0.70mil		
	Dielectric 3	ISOLA 370 HR	5.00mil	4.2	
3	SI61		0.70mil		
	Dielectric 4	ISOLA 370 HR	4.81mil	4.2	
4	GND2/PWR1		0.70mil		
	Dielectric 5	ISOLA 370 HR	5.00mil	4.2	
5	SI62		0.70mil		
	Dielectric 8	ISOLA 370 HR	4.81mil	4.2	
6	GND3/SIG3		0.70mil		
	Dielectric 10	ISOLA 370 HR	4.00mil	4.2	
7	GND4/SIG4		0.70mil		
	Dielectric 11	ISOLA 370 HR	4.81mil	4.2	
8	SI65		0.70mil		
	Dielectric 12	ISOLA 370 HR	5.00mil	4.2	
9	GND5/PWR2		0.70mil		
	Dielectric 13	ISOLA 370 HR	4.81mil	4.2	
10	SI66		0.70mil		
	Dielectric 14	ISOLA 370 HR	5.00mil	4.2	
11	GND6		0.70mil		
	Dielectric15	ISOLA 370 HR	6.05mil	4.2	
12	Bottom Layer		1.88mil		
	Bottom Solder	Solder Resist	2.00mil	3.5	
	Bottom Overlay				

DESIGN INFORMATION

MIN. TRACK WIDTH: 4 MIL
MIN. CLEARANCE: 4 MIL
MIN. VIA PAD SIZE: 16 MIL

MINIMUM ANNUAL RING 0.05mm (2MIL) EXTERNAL
PER IPC-D-275 CLASS 2 LEVEL C

REGISTRATION TOLERANCES: METAL +/- 5 MIL, HOLES +/- 3 MIL
HOLE SIZE TOLERANCE (UNLESS OTHERWISE SPECIFIED): +/- 3 MIL

MATERIAL:

☐ FR-408 ☐ FR-4 High Tg ☒ OTHER ISOLA-370 HR

THICKNESS: ☐ 62 MIL (1.6mm) +/-10% ☒ OTHER 1.7MM+/-10%

TOLERANCE: ☒ ANSI IPC-6012 TYPE 3 CLASS 2
☐ OTHER +/- _____

BOW & TWIST: ☒ ANSI IPC-6012 TYPE 3 CLASS 2
☐ OTHER +/- _____

DRILLING:

REFERENCE: ☒ AS SHOWN ☒ NC DRILL FILES
PTH COPPER THICKNESS: ☒ 20-30 um ☐ OTHER _____

BOARD FINISH:

SILKSCREEN: ☒ TOP ☒ BOTTOM
SILKSCREEN COLOR: ☒ WHITE ☐ OTHER _____

SOLDER RESIST COLOR: ☒ GREEN ☐ OTHER _____
☒ MATTE ☐ SEMI-GLOSS

SURFACE FINISH: ☒ IMMERSION GOLD (ENIG) ☐ ENEPIG
☐ IMM. TIN/SILVER OR EQUIV ☐ OTHER _____

ARRAY/PANEL: ☒ CUT AND TRIM PER M1 BOARD OUTLINE
☐ N.C. ROUTE ☐ V. SCORE

CERTIFICATION: MATERIALS AND WORKMANSHIP FOR ALL PCBs
TO MEET OR EXCEED THE REQUIREMENTS OF:

☒ ANSI IPC-A-600F CLASS -> ☐ 1 ☒ 2 ☐ 3
☒ RoHS ☐ OTHER PER ORDER

ALL BOARDS MUST MEET OR EXCEED UL94-V0 REQUIREMENTS.
PCB MUST BEAR THE UL94V-0 UL REGISTERED MATERIAL ID NUMBER

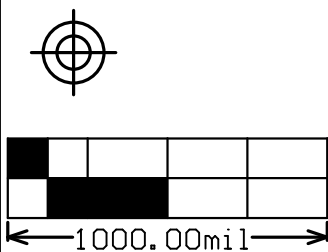
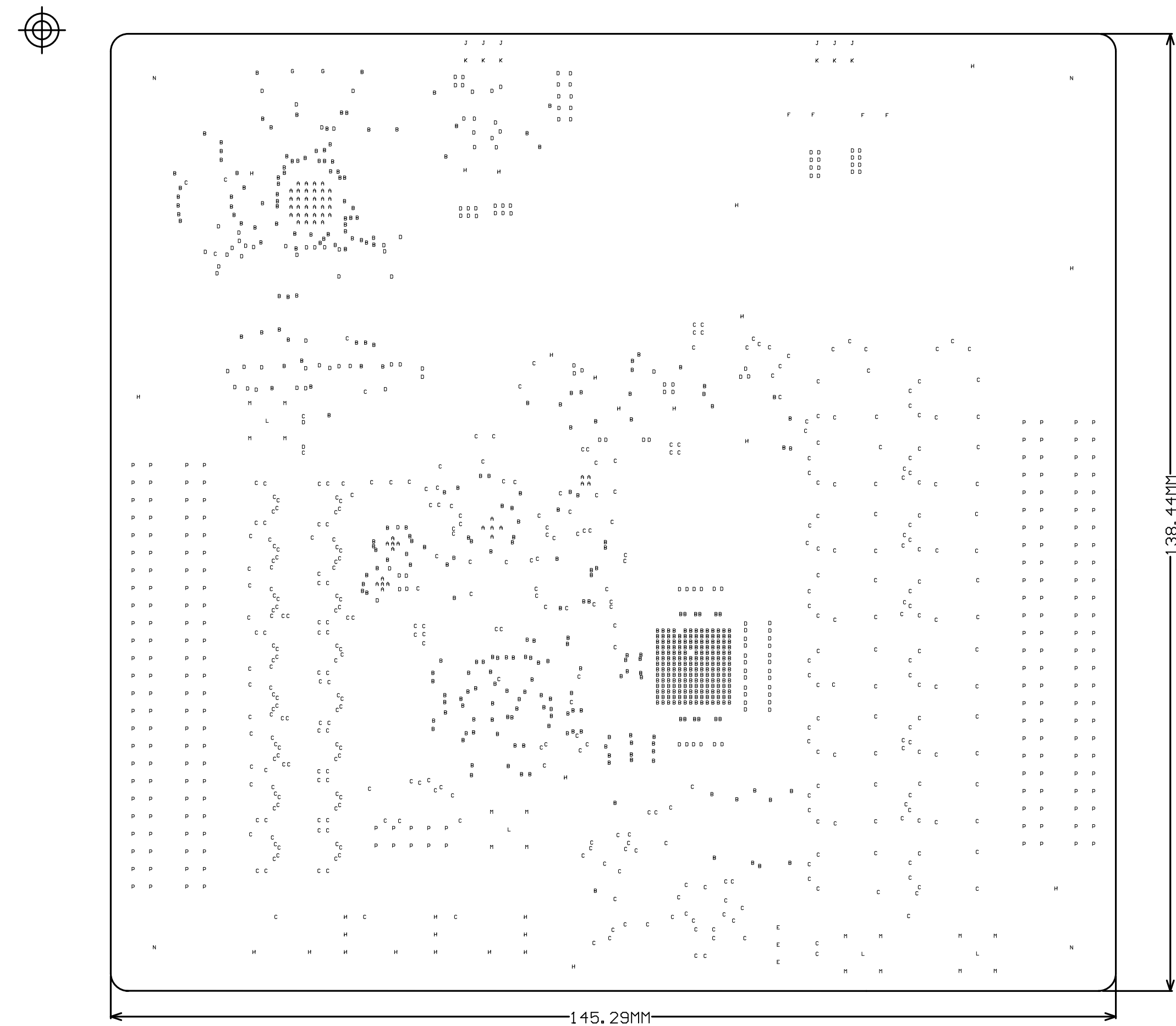
ADDITIONAL REQUIREMENTS:

MICROSECTION: ☐ YES

BARE BOARD ELEC. TEST: ☐ NONE ☒ REQUIRED ☐ PER ORDER

☐ XX MIL VIAS REQUIRE NON-CONDUCTIVE FILL AND PLANARIZE
☐ XX MIL VIAS REQUIRE CONDUCTIVE FILL AND PLANARIZE
☐ OUTER XX MIL TRACES REQUIRE 50 OHM SINGLE-ENDED IMPEDANCE
☐ LAYER 2 & 3 (INNER LAYERS) XX MIL WIDE, XX MIL SPACE
TRACES REQUIRE 100 OHM DIFFERENTIAL IMPEDANCE

	
PROJECT TITLE: TX7364 EUM	
DESIGNED FOR: Public Release	
FILE NAME: DC309A_PCB.PcbDoc	
ENGINEER: ROSHAN	LAYOUT BY: KAU IKANNAN
SCALE: 1.00	ALTUM DESIGNER VERSION: 24.1.2.44



ALL ARTWORK VIEWED FROM TOP SIDE	BOARD #: DC309	REV: A	SUN REV: Not in version control
LAYER NAME = M2 Solid Board Outline =Fabrication Drawing	TID #: N/A		
PLOT NAME =Fabrication Drawing	GENERATED : 2/2/2024	6:10:20 PM	TEXAS INSTRUMENTS

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